

Current solved papers

BT401 Final Term 8 AM 1st September (Aleeze Tahir Shah)

1. Briefly explain state of world's forest genetic resources. 3

The State of the World's Forest Genetic Resources

In 2014, the Food and Agriculture Organization of the United Nations published the first State of the World's Forest Genetic Resources. The publication addressed the conservation, management and sustainable use of forest tree and other woody plant genetic resources of actual and potential value for human well-being in the broad range of management systems.

2. What are three pillars of Ramsar Convention? 5

Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975

The “three pillars” of the convention

The Contracting Parties (160) commit to:

- Work towards the wise use of all their wetlands.
- Designate suitable wetlands for the list of Wetlands of International Importance.
- Cooperate internationally on transboundary wetlands, shared wetland systems and shared species.

3. What is the extreme weather effect on invertebrate's genetic resources? 5

Because of the many ecosystem services that they provide, invertebrates have a key role to play in adapting agriculture to the effects of climate change. The extent to which the individual services provided by invertebrates will be enhanced or impeded by climate change is difficult to predict. However, if invertebrate biodiversity is lost, the capacity of ecosystems to adapt is likely to diminish.

- Healthy soils – and healthy, diverse soil invertebrate communities – will be vital to climate change adaptation. For example, earthworms help to maintain soil structure and the availability of water throughout the soil profile. Studies have shown that the presence of these animals can help to alleviate the effects of drought on crop Production. Studies have also revealed the remarkable.

ability of diverse soil invertebrate communities to restore the structure of degraded soil. The potential for managing soil invertebrates to enhance their beneficial roles has been little explored. Few if any deliberate attempts have been made to introduce soil invertebrates into new countries or ecosystems. Given the potential for such species to become invasive, it is inadvisable to attempt any such introductions until soil ecology is much better understood than it is today. However, every effort should be made to avoid agricultural practices that disrupt resident soil invertebrate communities and the services they provide.

It is likely that some pests, as they move into new areas in response to climate change, will at least temporarily escape from their natural enemies. This is likely to increase demand for classical biological control agents, especially in places where the newly established pest population is separated from its original home by a physical barrier such as the sea or a mountain range. For this reason, access to new classical biological control agents is likely to be particularly important for island countries

4. Discuss the impact of climate change on FGR. Also discuss how this changing climate effect organism which is associated with trees? 10

Forest genetic resources are essential for forest-depending communities who rely for a substantial part of their livelihoods on timber and non-timber forest products (for example fruits, gums and resins) for food security, domestic use and income generation.

Forest genetic resources and climate change

Diversity of forest genetic resources enables the potential for a species (or a population) to adapt to climatic changes and related future challenges such as temperature changes, drought, pests, diseases and forest fires. Though forest trees are known for showing great plasticity in their response to climate changes, not all species are naturally capable to adapt at the pace necessary.

Climate change may also result in high variability in temperature and precipitation, with an increase in incidence of extreme events, such as flooding, late frosts and intensive summer droughts, amongst other events. In some areas, such as the Mediterranean and the Neo-tropics, an increase in seasonality is also expected. Under such conditions, natural selection may not result in efficient adaptation because selection pressures are multi-directional, involving traits that may be inversely correlated at the gene level. The standing genetic variation in populations may then not be large enough to create the rare new genotypic combinations that are required. Ecosystems affected by abrupt change may sustain rapid and widespread transformation as ecological tipping points are exceeded. Given the pivotal role of trees in ecosystem function, abrupt climate change impacts on them may thus have profound consequences for forests as a whole. Irreversible loss of ecosystem integrity and function may follow, with replacement by new nonendemic ecosystems.

Direct impacts of climate change

These include high tree mortality through extreme climatic events, particularly drought in combination with widespread regeneration failure, for example, examined the evidence for anthropogenic climate change leading to future large-scale “dieback” in Amazonian rain forest. Analysis suggested that dryseason water stress is likely to increase in eastern Amazonia over the 21st century, with the region tending toward a climate more appropriate to seasonal forests.

Effects of changing climate on organisms associated with trees

In particular, changes in the biology of insect pests and diseases may make ecosystems more susceptible to tree mortality. Because of improved environmental conditions for the pest and reduced tree resistance due to increased stress, pests may react to climate change with range expansions and/or increases in attack severity.

Changes in abiotic disturbance regimes:

These include changes in fire regimes, flooding, landslides and/or hurricanes. Fire and climate are closely linked and are also associated with changes in land use. Coupled climate and fire-risk models suggest not only an increase in the frequency of fires but also in fire size and length of the fire-risk season, with some areas subject to risk that were not before. **Malhi et al. (2009)** considered how tipping points may be reached in Amazonian rainforest by a combination of increased dryness and an increased incidence of fire events

Invasion by organisms foreign to local ecosystems

This includes the increased risk of establishment by invasive species which accidentally arrive into ports of entry, through globalized commerce. By making new niches available, climate change will facilitate the survival of mammals, insects, diseases and/or weeds foreign to endemic ecosystems.

5. What types of services are provided by PEPA? 5

Pakistan Environmental Protection Act, 1997

PEPA provides for;

- Protection,
- Conservation,
- Rehabilitation and
- Improvement of the environment;
- PEPA provide framework for prevention and control of pollution
- Helps in protection of sustainable development.

6. What areas are covered by Cartagena protocol?

The Bio safety (Protocol to CBD) deals with;

- | |
|---|
| <ul style="list-style-type: none">▪ Safe handling,▪ Storage▪ Trans-boundary movement of the Genetically Modified Organisms (GMO).▪ |
|---|

Cartagena protocol was adopted on June 2001 in Cartagena, Spain. It entered into force on September 11th; 2003. Pakistan signed the Cartagena protocol in June 2000. Pakistan has ratified it in May 2009

Objectives of Cartagena Protocol on Biosafety

The Protocol states that it aims to;

- Contribute to ensuring an adequate level of protection in the field of the safe transfer, handling and use of living modified organisms resulting from modern biotechnology that may have adverse effects on the conservation and sustainable use of biological diversity,
- It takes into account risks to human health, and specifically focusing on transboundary movements.
- It seeks to protect biodiversity from the potential risks of living modified organisms (LMOs) resulting from modern biotechnology.

Cartagena Protocol areas

The Protocol covers:

- Transboundary movement, transit, handling and use of all living modified organisms that may

have adverse effects on the conservation and sustainable use of biological diversity, taking into account risks to human health.

It does not cover:

- Products derived from LMOs (e.g. paper from GM trees)
- LMOs, which are pharmaceuticals for humans that are addressed by other relevant international agreements or organizations

7. Briefly explain historical background of Ramsar convention. 3

History: Ramsar is one of the global inter-governmental environmental agreements. The treaty was negotiated in 1960s by countries and NGOs. To avoid the increasing loss and degradation of wetland habitat for migratory water birds. In an 18 nations meeting it was adopted in the Iranian city of Ramsar On 2nd February 1971. Came into force in 21st December 1975

8. Describe purpose of panel code. 2

Pakistan Panel Code, 1860

The polluter of the environment can be punished under this code for certain types of pollution.

These punishment are of following types;

- Punishment for water pollution
- Punishment for atmospheric pollution
- Punishments for general pollution

9. What is location of cholistan wildlife sanctuary? 2

Cholistan Wildlife Sanctuary

It is part of the Cholistan desert in the south eastern portion of the province of Punjab. It contains some of the most rare and interesting wildlife in Pakistan. Some of the rare animals of this region are Desert wolf (rare), Indian fox, Red fox, Jackal, Small Indian civet, Small Indian mongoose, Indian grey mongoose, Indian desert cat, Jungle cat, Caracal cat, Saker falcon, Black backed vulture, Indian cobra, Monitor lizard, Saw scaled viper and Russells viper.

10. Mission of Ramsar convention.

Mission

The Convention's mission is

“The conservation and wise use of all wetlands through local and national actions and international cooperation, as a contribution towards achieving sustainable development throughout the world.

11. MCQS were easy from mid and quiz

Bt401 final paper spring 2018

Total Q 50 (80 marks)

40 Mcqs 6 7 mcqs were from last quiz

6 short Q (2 and 3 marks)

1. What is botanical garden? (2)

It is a garden dedicated to the collection, cultivation and display of a wide range of plants labeled with their botanical names. It may contain specialist plant collections such as tropical plants, or other species of plants.

Types of plants in botanical gardens

1. cacti and succulent plants.
2. herb gardens.
3. greenhouses, shade houses

2. Define forest genetic resources? (2)

Forest genetic resources are essential for forest-dependent communities who rely for a substantial part of their livelihoods on timber and non-timber forest products (for example fruits, gums and resins) for food security, domestic use and income generation.

3. On how many tinypoly tress rely on maintain the climate? (2) asy h kuch tha Q yad ni

Tree populations rely on three interplaying mechanisms to respond to environmental change:

1. adaptation
2. Migration
3. phenotypic plasticity

4. Write the variations in variation gene? (3)

There are three primary sources of genetic variation, which we will learn more about:

- Mutations are changes in the DNA. ...
- Gene flow is any movement of genes from one population to another and is an important source of genetic variation.
- Sex can introduce new gene combinations into a population.

5. Define strict reserve conviction? (3)

6. Define seed storage? (3)

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed too readily as a basic tool in maintaining genetic diversity. Conventional seed storage offers several advantages:

1. Seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation
2. Good storage facilities are now available in most of the world, and they are used extensively for

tree seed storage for various regeneration purposes;

3. Seed storage is a relatively cheap method for conserving a broad range of germplasm

4. Large land areas are not tied up in conservation

5. International exchange of genetic material is facilitated by seed storage.

4 long Q (5 and 10 marks)

1. Write features of cartagona protocol? (5)

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2. Write note on convention of botanical garden? (5)

Botanical Garden

A botanical garden or botanic garden is a garden dedicated to the collection, cultivation and display of a wide range of plants labeled with their botanical names.

Types of plants in botanical gardens

1. cacti and succulent plants.

2. herb gardens.
3. greenhouses, shade houses.
4. tropical plants.
5. Medicinal Plants.
6. aromatic or textile plants
7. other exotic plants.

Who is responsible for a botanical gardens?

Botanical gardens are often run by universities or other scientific research organizations, and often have associated herbaria and research programmers in plant taxonomy or some other aspect of botanical science.

Importance of Botanical garden

1- Enjoyment

2- Economic

3- Scientific research

Botanic gardens contain collections of plants for education, scientific purposes and display; they can be: taxonomically-based - collections of a particular family, genus or group of cultivars; or collections of native plants; or useful species such as medicinal, aromatic or textile plants.

4- Conservation

Conservation of rare and threatened plants. The conservation of plant diversity is critical for sustainable development and botanic gardens are playing a key role as centers of conservation action. Botanical gardens can promote diversity. Because they include many species of plant.

5- Climate Change

Plants can alter the temperature of the Earth's atmosphere. Through the process of photosynthesis, plants use energy from the sun to draw down carbon dioxide from the atmosphere and then use it to create the carbohydrates they need to grow. Since carbon dioxide is one of the most abundant greenhouse gases, the removal of the gas from the atmosphere may temper the warming of our planet as a whole. *transpiration in plants can increase water vapor in the atmosphere, causing more precipitation and cloud cover in an area. The additional cloud cover often reinforces the cooling by blocking sunlight. *Contribute to soil fertility and prevent soil erosion.

3. What is cold storage. Discuss? (5)

Cold storage - it involves storage in low and non freezing temperature.

Slow Freezing- Slow thawing

With this method organs are labeled into vials after equilibration with a cryoprotectant solution and then cooled at rate of 0.5-2 °C per minute down to -1 °C. Seeding is then induced and a holding period of 5 to 15 minutes allows equilibration of the temperature. Thereafter embryos are cooled to -60 °C or lower at a rate of 0.3 to 0.5 °C per minute before being transfer to liquid nitrogen. Frozen embryos must be slowly thawed at a rate of less then 25 °C per minute to prevent osmotic shock

2. Rapid cooling and rapid thawing

In this technique , however cooling is terminated at -30 to -40 °C and embryos are then plunged into liquid nitrogen for rapid cooling to -196 °C. Thawing is therefore performed rapidly (200 to 500 °C per

minute) to prevent recrystallization

3. Vitrification

Vitrification is the process of cooling where the water in the tissue becomes glass rather than crystals. Glass is a liquid that is too cold (too viscous) to flow. In other words vitrification is solidification due to increased viscosity rather than crystallization.

4. Ultrarapid Freezing

In this technique serial equilibration of embryos in high concentration of DMSO (3-5 M) supplemented with sucrose (0.3 to 0.5 M). The embryos are then plunged into liquid Nitrogen. Thawing is then done with warm water bath (approximately 500 °C per minute).

4. Write the impact of climate change on FGR. Also discuss the climate change on organisms which associate with FGR? (10)

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BT401 Today Paper

40 mcq thy

Define Treaty (2)

Define Landscape and Seascape protected area (2)rep

What is Cholistan protected area name (2)rep

Write the objectives of Mounment features (3)

Primary objective

To protect specific outstanding natural features. Their associated biodiversity and habitats

Other objectives

- To provide biodiversity protection in landscapes or seascapes that have otherwise undergone major changes
- To conserve traditional spiritual and cultural values of the site.

Write the objective of socio asa hi tha kuch 5 marks ka

What is cold storage. Discuss? (5)rep

Write the impact of climate change on FGR. Also discuss the climate change on organisms which associate with FGR? (10)rep

Write the objectives of CBD (10)

Convention on Biological Diversity, 1992

•	CBD is about the conservation and wise use of different biological resources (plants and
animals).	
	It was adopted in 1992 at Rio De Janeiro, Brazil and entered force on January, 1993, which was 90 days after the 30th ratification.

	Pakistan signed it in June 1992 at United Nations Conference on Environment and Development held at Rio De Janeiro, Brazil Pakistan ratified it on 26th July 1994. The Convention on Biological Diversity covers biodiversity at all levels:
	Ecosystems, Species
	Genetic resources
	It also covers biotechnology, including through the Cartagena Protocol on Biosafety. In fact, it covers all possible domains that are directly or indirectly related to biodiversity and its role in development, ranging from science, politics and education to agriculture, business, culture
and much more.	

Objectives of CBD, 1992

The three inter-related objectives are:

• The conservation of biological diversity; • The sustainable use of its components;	
The fair and equitable sharing of the benefits arising out of the utilization of genetic resources, including by appropriate	
• Access to genetic resources, • Transfer of relevant technologies, • Funding.	

Facts & Figures of CBD, 1992

According to CBD;

- Ecosystems, species and genetic resources should be used for the benefit of humans, but in a way that does not lead to the decline of biodiversity.
- Substantial investments are required to conserve biodiversity, but it will bring significant environmental, economic and social benefits in return.
- The Ecosystem Approach, an integrated strategy for the management of resources, is the framework for action under the Convention.
- The precautionary principle states that where there is a threat of significant reduction or loss of biological diversity, lack of full scientific certainty should not be used as a reason for postponing measures to avoid or minimize such a threat.

bt 401

1. imitations of germplasm conservation 5

Germplasm:

Germplasm of a crop may be define as the sum total of all hereditary material. i.e. all the alleles

of various genes, present in a crop species and its wild and weedy relatives. It is also termed as

Genetic Resources.

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed

too readily as a basic tool in maintaining genetic diversity. Conventional seed storage offers several

advantages:

1. Seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation
2. Good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes;
3. Seed storage is a relatively cheap method for conserving a broad range of germplasm
4. Large land areas are not tied up in conservation
5. International exchange of genetic material is facilitated by seed storage.

Limitations

1. seeds of many important tree species cannot adequately survive long-term storage;
2. seeds of many tropical hardwoods cannot survive even a year of conventional seed storage;
3. some genetic damage or change in gene frequencies may occur during seed storage. Cryogenic storage at its current level of technology offers great promise for long-term storage of true orthodox and sub-orthodox seeds. It has potential for application to recalcitrant species, but much more research and development will be required. Some advantages of cryogenic storage are:

	it can apparently extend storage life of true-orthodox species far beyond that possible in conventional storage; genetic damage may be much less than what occurs in conventional storage.
•	Disadvantages include special equipment is required, which is not readily found in forestry facilities; (2) it is costly for large seeds recalcitrant seeds (both classes) cannot be stored cryogenically with current technology.

2. climate change of FGR wala tha ak question 10 marks ka rep

3. plant genetic resources institute names. 2

These institutions

include

- Ministry of Food
- Agriculture and Livestock at federal level
- livestock departments at provincial level
- The main institutes devoted to research for the development of livestock are: Animal

- Sciences Institute (ASI) at National Agricultural Research Centre, Islamabad;
- Livestock Production Research Institute (LPRI), Bahadurnagar, Okara;
- Barani Livestock Production Research Institute (BLPRI), Kherimurat, District Attock;
- Research Institute for physiology of Animal Reproduction (RIPAR), Bhunikey, Pattoki,
- Kasur; Poultry Research Institute (PRI), Rawalpindi;
- Animal Nutrition Research Centre, Rakh Dera Chahl, Lahore and Poultry Research
- Institute, Karachi

4. name physical method for storage

There are various methods of storage :

1. **Cryopreservation** - generally involves storage in liquid nitrogen.
2. **Cold storage** - it involves storage in low and non freezing temperature.
3. **Low pressure** – it involves partially reducing the atmospheric pressure of surrounding.
4. **Low oxygen storage** - it involves reducing the oxygen level but maintaining the pressure.

5. primary objectives of national park

Primary objective

To protect natural biodiversity along with its underlying ecological structure.

Other objectives

- To manage the area in order to perpetuate, in as natural a state as possible.
- To maintain viable and ecologically functional populations.
- To contribute to local economies through tourism.
- To manage visitor use for inspirational, educational recreational purposes.

6. runn of kutch wildlife sanctuary

Runn of Kutch Wildlife Sanctuary

It spread over 566,375 ha is part of the great Thar desert and comprises. Rann of Kutch across the frontier with India, which includes permanent saline marshes, coastal brackish lagoons, tidal mudflats, and estuarine habitats. The site supports many locally and globally threatened species, Threatened species include the Great Indian bustard, Houbara bustard, Sarus crane this area used to have the only population of the Indian Wild Ass or Onager in Pakistan.

7.phenotypic plasticity

The role of phenotypic plasticity

Phenotypic plasticity is defined as the capacity of a particular genotype to express different phenotypes under different environmental conditions. The concept is often extended to populations and species. with plastic trees those with flexible morphology and physiology that grow at least reasonably well under a range of different environmental stresses without genetic change. A degree of phenotypic plasticity is found in most trees, but varies substantially amongst and within species. Even in species with very little

genetic diversity, such as *Pinus pinea* L., strong phenotypic plasticity is expressed for growth-related traits, which may have helped the species colonise new environments. At least in the short term, high plasticity is likely to favour tree survival under changing environmental conditions, although trade-offs between traits can be expected. Since phenotypic plasticity has a heritable basis and may be selected for under changing environments, complex interactions between traits are possible, depending on the magnitude and structure of change.